CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (Civil) Examination
May 2011

CL 211 Environmental Engineering I

Date: 13.05. 2011, Thursday | **Time: 10:00 a.m. To 01:00 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Write short note on: "Water Borne Diseases". **03**

(b) Explain the role of Environmental Engineer. **04**

Q - 2 (a) List the physical characteristics of water. Explain any two in details. **03**

(b) Draw the layout of surface water treatment plant. Mention the function of each unit. **04**

(c) What is discrete particle settling? Explain the discrete particle settling. **03**

(d) What is coagulation? Explain the theory of filtration. **04**

OR

Q - 2 (a) Differentiate between Slow Sand filter and Rapid Sand Filter. **03**

(b) Define the term sedimentation. Why it is necessary for water treatment? **04**

(c) Explain "Break Point Chlorination". **03**

(d) Compare the characteristics of different coagulants used in water treatment. **04**

Q - 3 **Attempt any Two:** **14**

(a) Design a circular sedimentation tank filled with mechanical sludge remover for water works which has to supply daily 3.5 million liters of water to the town. The detention period in the tank for water for 4 hours and depth of water in the tank may be assumed as 3.5m.

(b) What is hardness? Give classification of hardness. Describe the method for removal of hardness by lime-soda process.

(c) Sketch and describe in detail the working of Pressure filter.

SECTION – II

Q - 4 (a) What is potable water? What are the requirements of wholesome water? **03**

(b) What is the formula used for the firefighting demand? Explain the various factors affecting the per capita demand. **04**

Q - 5 (a) What is the purpose of intake structure? Draw a neat sketch of canal intake. **03**

(b) Draw the neat sketch showing the house service connection from the distribution main and state the function of water main, ferrule and service pipes. **04**

(c) What is intermittent system of water supply? List advantages of P.V.C. pipes over **03**

steel pipes for water supply.

(d) Define 'Ventilation'. Explain gully trap in detail. **04**

OR

Q – 5 (a) What are points should be kept in mind while selecting a site for intake structure? What is meant by per capita demand? **03**

(b) Draw the neat sketch of overhead tank and state the function of each component. **04**

(c) What are the requirements of a distribution system? List any three requirements of pipe material to convey water. **03**

(d) Enlist systems of ventilation. Explain balanced system of ventilation. **04**

Q – 6 **Attempt any Two.** **14**

(a) Explain the methods of refuse disposal.

(b) Explain different sanitary fittings in house drainage system.

(c) Give importance of air conditioning and lighting in a ventilation system.

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